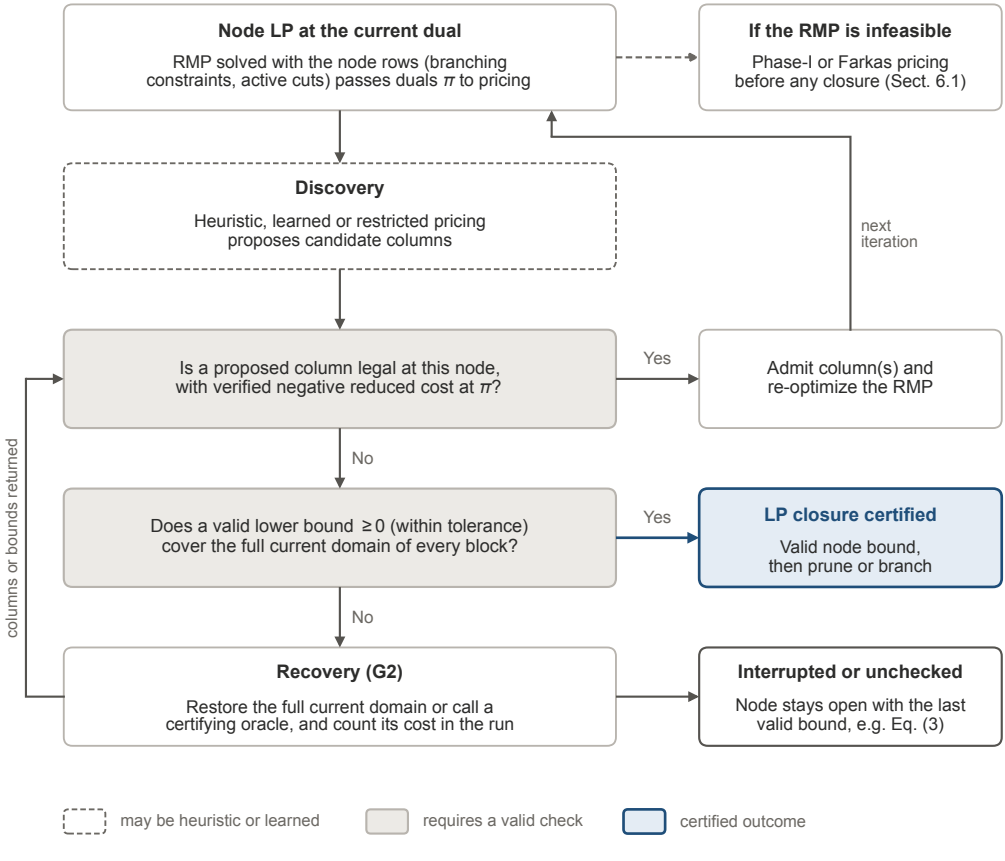
Node LP at the current dual
RMP solved with the node rows (branching constraints, active cuts) passes duals π to pricing
If the RMP is infeasible
Phase-I or Farkas pricing before any closure (Sect. 6.1)
Discovery
Heuristic, learned or restricted pricing proposes candidate columns
next iteration
Is a proposed column legal at this node, with verified negative reduced cost at π?
Yes
Admit column(s) and re-optimize the RMP
No
columns or bounds returned
Does a valid lower bound ≥ 0 (within tolerance) cover the full current domain of every block?
Yes
LP closure certified
Valid node bound, then prune or branch
No
Recovery (G2)
Restore the full current domain or call a certifying oracle, and count its cost in the run
Interrupted or unchecked
Node stays open with the last valid bound, e.g. Eq. (3)
may be heuristic or learned
requires a valid check
certified outcome